



**Submission Title:** Smart Digital Therapeutics for Substance Use Disorders: AI Algorithms for Daily Recovery Health Prediction and Personalized Micro-interventions

## SUBMISSION PREVIEW: SMART DIGITAL THERAPEUTICS FOR SUBSTANCE USE DISORDERS: AI ALGORITHMS FOR DAILY RECOVERY HEALTH PREDICTION AND PERSONALIZED MICRO-INTERVENTIONS

[Smart Digital Therapeutics for Substance Use Disorders: AI Algorithms for Daily Recovery Health Prediction and Personalized Micro-interventions](#)

Submission ID: 2483300

Submission Category: West Coast Symposium 2027

Submission Status: Active

### Presenter(s)

**John J. Curtin, PhD (he/him/his)**

**Position:**

Professor

**Organization:**

University of Wisconsin - Madison

**Role:**

Presenter

**Bio**

John J. Curtin, PhD

Professor Curtin is a full professor in the Department of Psychology at the University of Wisconsin-Madison. He is the Director of the Addiction Research Center (<https://arc.psych.wisc.edu/>) and the former Director of Clinical Training of the doctoral program in Clinical Psychology in this department. He also directs the Data Science in Human Behavior master's program in Psychology. He is the co-founder and Chief Science Officer at RudraHealth AI (<https://www.rudrahealth.ai/>), which develops digital therapeutics for substance use disorder.

Professor Curtin's program of research uses personal sensing and machine learning techniques for psychiatric diagnosis and risk prediction, precision mental health, and "just-in-time" interventions for substance use and other psychiatric disorders. His scholarship has been recognized by numerous grants from the National Institutes of Health (NIAAA and NIDA) and awards from the American Psychological Association, the Association for Psychological Science and UW-Madison. Professor Curtin also contributes to UW-Madison's teaching mission, offering courses on substance use disorders, statistics, and applied machine learning.

**Have you presented at other educational conferences within the last 2 years?**

Yes

**If yes, where?**

- Curtin, JJ (2026, March). Panelist in "AI, Ethics, and the Humanities" round table hosted by the Stephen and Laurel Brown Foundation at the Upper House Commons, Madison, WI. - Curtin, JJ (2026, March). "Smart" Recovery Monitoring and Support for SUD Continuing Care. Presented at the Fun with Affective Neuroscience meeting, Center for Healthy Minds, University of Wisconsin, Madison, WI - Curtin, JJ (2025, October). "Smart" Recovery Monitoring and Support for SUD Continuing Care. Presented the Letters and Science Board of Visitors meeting, University of Wisconsin, Madison, WI - Curtin, JJ (2025, April). "Smart" Recovery Monitoring and Support for SUD Continuing Care. Presented at the monthly meeting of the Neurotech Society. University of Wisconsin, Madison, WI - Punturieri CE, Wanta SE, Curtin, JJ (2025 April). Performance and Equity of Geolocation Data for Lapse Prediction in Alcohol Use Disorder. Presented at the Annual Meeting of the Society of Biological Psychiatry, Toronto, Ontario, Canada. - Punturieri CE, Wanta SE, Curtin, JJ (2025 April). Performance and Equity of Geolocation Data for Lapse Prediction in Alcohol Use Disorder. Presented at the 13th Annual Collaborative Perspectives on Addiction Meeting, Providence, Rhode Island. - Curtin, JJ (2024, December). "Smart" Recovery Monitoring and Support for SUD Continuing Care. Presented at the 63rd Annual Meeting of the American College of Neuropsychopharmacology, Phoenix, AZ. - Curtin, JJ (2024, November). Developing a "Smart" Recovery Monitoring and Support System. Presented at the University of Wisconsin-Madison Emotion Training Grant Seminar, Madison, WI. - Curtin, JJ (2024, November). A "Smart" Recovery Monitoring and Support System for SUD Continuing Care. Presented at the Clinical Area Group "Meet the Faculty", University of Wisconsin, Madison, WI. - Curtin, JJ (2024, October). A "Smart" Recovery Monitoring and Support System for SUD Continuing Care. Presented at: University of Wisconsin-Madison Department of Psychology Board of Visitors Meeting, Madison, WI. - Curtin, JJ (2024, October). Developing a "Smart" Recovery Monitoring and Support System. Presented at Center for Healthy Minds Research Meeting, Madison, WI. - Wyant KM Sant'Ana SJ, Fronk GE & Curtin JJ. (2024, June). Using ecological momentary assessment for temporally precise lapse prediction in alcohol use disorder. Flash talk presented at the

2024 Society for Ambulatory Assessment Conference, Ann Arbor, MI. - Curtin, JJ (2024, May). Panelist in roundtable on "How to Measure and Model Recovery." Hosted by the NIH HEAL Initiative Workshop: Substance Use Disorder Trajectories and Outcomes Within the Broader Socio-Environmental Context, Washington, DC. - Curtin, JJ (2024, May). Precision Mental Health for SUD: Contributions from Sensing and Algorithms. Presented at University of Washington Cross-Area Clinical Seminar, Seattle, WA. - Fronk GE, Papp A, Wyant K, Sant'Ana SJ, & Curtin JJ. (2024, April). Dynamic assessment of self-efficacy predicts future alcohol use. 12th annual meeting of Collaborative Perspectives on Addiction, Denver, CO. - Yu J, Wyant K, Sant'Ana SJ, Fronk GE & Curtin JJ. (2024, April). Assessing demographic variability in machine learning model performance to predict alcohol lapses. 12th annual meeting of Collaborative Perspectives on Addiction, Denver, CO. - Curtin, JJ (2024, March). Precision Mental Health for SUD: Contributions from High Dimensional Prediction Models. Presented at the Clinical Area Group "Meet the Faculty", University of Wisconsin, Madison, WI. - Curtin, JJ (2024, March). Smart Digital Therapeutics for Alcohol Use Disorder: Algorithms for Prediction and Adaptive Intervention. Presented at DSHB Research Meeting. - Curtin, JJ (2025, January). Panelist in roundtable on "Data ethics and algorithmic bias in mental health research." Hosted by DrivenData and the Wellcome Trust's Data for Science and Health Team, Denver, CO.

CV

cv.pdf

### Presenter Qualification Category

Category 1 - Hold a graduate degree IN A MENTAL HEALTH FIELD and be qualified by appropriate education experience and training to present the particular subject matter

### Professional Licenses and Certifications

PhD in Clinical Psychology

### Who will be responsible for the faculty's expenses

List my company as a supporter - it will be responsible for my expenses

### Please list the organization that will be supporting you

RudraHealth AI

## Submission

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### Topic Area

1st choice: Relapse Prevention/Aftercare

2nd choice: AI Tools

### Has this topic been, or will it soon be, presented at any other industry meetings?

- NO

### Did you present this content at a previous event?

- No

## Level of Instruction

- Advanced

## Presentation Length

- 60 or 90-Minute Plenary/Keynote
- 90-Minute Breakout Session

## Session Description

### OBJECTIVES:

- Describe current and near-term future uses of digital therapeutics for substance use disorders (SUD).
- Understand how active and passive personal sensing can be used to monitor patients' recovery from SUD
- Understand how personal sensing measures can be combined with machine learning to predict a return to goal-inconsistent substance use among individuals in SUD recovery.
- Describe how these machine learning models can be used to improve patient use of digital therapeutics by recommending optimal daily micro-interventions and/or supportive activities to enhance their recovery health

### PRESENTATION OUTLINE:

- Describe methods for active and passive sensing during early recovery from substance use disorders, including alcohol use disorder and opioid use disorder.
- Present results from two large NIH funded trials that developed and evaluated machine learning models to predict lapses back to substance use among patients in recovery from either alcohol use disorder (study 1) or opioid use disorder (study 2)
- Explicitly evaluate the algorithmic fairness of these models when used with subgroups that have been historically under-served and/or displayed disparities in SUD treatment outcomes.
- Use tools from interpretable AI to look "under the hood" of these algorithms to understand what they tell us about the proximal risk factors for substance use lapses.
- Discuss plans to use these models to provide daily, algorithm-guided recommendations to personalize and adapt digital therapeutic use for patients with SUD.

Session participants will learn about current opportunities provided by emerging "smart" digital therapeutics that incorporate AI.

## Scientific Basis or Research

We present the methods for personal sensing and AI/machine learning algorithms that can be used to predict a return to goal-inconsistent use (e.g., lapse) among individuals with substance use disorders. These same algorithms can be used to make personalized daily recommendations for micro-interventions and/or other recovery support activities each day. We discuss how these algorithms can be embedded within a digital therapeutic system to provide scalable, long-term continuing care for individuals in recovery.

Pulick E, Curtin J, Mintz Y (2025). Idiographic Lapse Prediction With State Space Modeling: Algorithm Development and Validation Study. *JMIR Form Res*, 9:e73265. DOI: 10.2196/73265 Open access

Wyant K, Sant'Ana SJK, Fronk G, & Curtin, JJ (2024). Machine learning models for temporally precise lapse prediction in alcohol use disorder. *Journal of Psychopathology and Clinical Science*, 133(7), 527-540

Wyant K, Moshontz H, Ward SB, Fronk GE, Curtin JJ (2023). Acceptability of Personal Sensing Among People With Alcohol Use Disorder: Observational Study. JMIR mHealth and uHealth, 11:e41833

## Bibliographic References

Wyant K, Sant'Ana S, Punturieri C, Yu J, Fronk G, Maggard C, Janssen C, Wanta S, Kornfield R, van Swol L, Curtin J (2025). Maximizing Engagement, Trust, and Clinical Benefit of AI-Generated Recovery Support Messages for Alcohol Use Disorder: Protocol for an Optimization Study. JMIR Res Protoc, 14:e81697

Pulick E, Curtin J, Mintz Y (2025). Idiographic Lapse Prediction With State Space Modeling: Algorithm Development and Validation Study. JMIR Form Res, 9:e73265. DOI: 10.2196/73265 Open access

Wyant K, Sant'Ana SJK, Fronk G, & Curtin, JJ (2024). Machine learning models for temporally precise lapse prediction in alcohol use disorder. Journal of Psychopathology and Clinical Science, 133(7), 527–540

## Does your session require a specific room set?

No

## Learning Objective

1. Upon completion, participant will be able to describe how personal sensing methods can be used to provide highly granular and temporally precise monitoring of real-world substance use relevant experiences
2. Upon completion, participant will understand the opportunities that emerging digital therapeutics provide to augment traditional clinician-delivered care
3. Upon completion, participant will recognize the key methods and dimensions to evaluate when implementing AI for clinical care including - overall performance, score calibration, algorithmic fairness among subgroups, explanatory benefits.